BACHELOR IN DATA AND BUSINESS ANALYTICS

TIME SERIES ANALYSIS

Session 1 · Introduction & Course Setup

Professor: Angel Mora Segura

02 September 2026 · 18:00–19:30 · Live in-person





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YOUR PROFESSOR

Ángel Mora Segura

Data & AI Architect at Nationale-Nederlanden (NN) · Ph.D. in Computer Science

- Extensive industry experience as an AI Architect, including roles at SDG Group and Nationale-Nederlanden, building data and AI systems for large organizations.
- Ph.D. researcher in software and model engineering at Universidad Autónoma de Madrid, with publications on modelling assistants and model transformation languages.
- Teaching experience across Computer Science and Business programs, combining academic rigor with real-world data science practice.

COURSE SCHEDULE · 20 SESSIONS · 80 MINUTES EACH

Session by Session

1	02 Sep	Introduction · R setup · forecasting as a business problem **TODAY**
2	03 Sep	Time plots, seasonal plots, EDA
3	23 Sep	Lags, ACF, PACF
4	24 Sep	Decomposition, STL, Box-Cox
5	30 Sep	Benchmarks: Naive & Seasonal Naive
6	01 Oct	Simple exponential smoothing + Holt
7	07 Oct	Holt-Winters / ETS
8	08 Oct	ETS model selection + validation
9	15 Oct	**Intermediate Test — 20% · Checkpoint 1**
10	19 Oct	AR / MA
11	22 Oct	ARMA + stationarity
12	28 Oct	ARMA + integration + diagnostics
13	29 Oct	ARIMA
14	12 Nov	Seasonal ARIMA (SARIMA)
15	18 Nov	Automatic ARIMA + model selection
16	19 Nov	Complex seasonality / Fourier · Checkpoint 2
17	25 Nov	Integrated Forecasting Workshop
18	26 Nov	Business Case review & feedback
19	03 Dec	**Group presentations — 30%**
20	11 Dec	**Final exam in R — 40%**

HOW THIS COURSE WORKS

You Are an Advanced Analytics Team in a Consulting Firm

From today, treat this course as the job. Your team of 3–4 is the advanced analytics unit of a consultancy, and you have just been assigned a client engagement: a real company with a real forecasting problem at the centre of a business decision.

"You are the analytics team responsible for addressing the forecasting problem described in your assigned business case. The case is your starting point, not a complete specification."

YOUR CLIENT
The company in your case

THE ENGAGEMENT
One semester of consulting work

THE DELIVERABLE
A forecasting approach you can defend in front of the client

An Applied Course

Time Series Analysis progressively connects forecasting methodologies with real business problems. It combines three dimensions, and you are assessed on all three.

01

Technical foundations

Understanding and applying the main time series forecasting methodologies covered in the course.

02

Practical implementation

Implementing, evaluating and comparing forecasting approaches in R.

03

Business application

Understanding how forecasting supports decision-making across industries and business contexts.

No technique is taught as an isolated topic. The objective is not to learn how to fit models, but to understand when, why and how each approach should be used.

BY THE END OF THE COURSE YOU SHOULD BE ABLE TO

Learning Objectives

- **Identify and interpret** the main components and patterns of a time series: trend, seasonality and autocorrelation.
- **Apply** Exponential Smoothing, AR/MA, ARMA, ARIMA and Seasonal ARIMA models.
- **Recognise complex seasonality** and determine an appropriate modelling strategy for it.
- **Compare alternatives** and justify model-selection decisions.
- **Build baseline models** and use them as the reference against which more advanced approaches are judged.
- **Evaluate models** using appropriate validation strategies, diagnostic tools and performance metrics.
- **Implement** every methodology in the course using R.
- **Translate results** into meaningful business insights and recommendations.

And above all: recognise that **the most sophisticated model is not necessarily the most appropriate one** for a particular business problem.

FROM METHODOLOGY TO BUSINESS APPLICATION

Every Technique Passes Through Your Case

Nothing is learned in the abstract. Each methodology introduced in class completes the same four-step loop before the next one arrives.

LEARN

Understand the concept and its implementation in R.

- What problem does it solve, what does it assume, and when does it break?
- Run it on the shared dataset until the code is yours, not mine.

→

APPLY

Evaluate whether and how it contributes to your business problem.

- Fit it to your own series, or argue on evidence why it does not apply.
- Record the result against your baseline — every session adds a line to the log.

→

DISCUSS

Share approaches, assumptions, results and difficulties.

- Case Clinic: 5 minutes on what you tried, found, or got stuck on.
- Four different cases means you see the same method behave four ways.

→

REFINE

Incorporate feedback and develop the final solution.

- Revisit earlier decisions when new evidence contradicts them.
- The recommendation gets sharper each week instead of being written at the end.

TYPICAL CLASS STRUCTURE · 80 MINUTES

The Classroom Is the Laboratory

The same three blocks every session. The work on your case happens here, with me in the room.

~30 MIN

BLOCK 1

Concepts & R application

The methodology, its intuition and its assumptions — aimed at **what problem this technique solves** — then straight into its practical implementation in R.

You leave with code that runs.

~30 MIN

BLOCK 2

Business Case Lab

Teams apply or evaluate the methodology in the context of their own case. Finished code is not the requirement.

Output can be a plot, a methodological decision, a hypothesis, a data transformation, a fitted model, or a justified argument for why this technique does not apply.

~20 MIN

BLOCK 3

Business Case Clinic

Selected teams share their approach, findings or difficulties in about 5 minutes each.

The class discusses alternatives and gives feedback; I close the loop.

Block durations vary with the topic: technically demanding sessions take more time for explanation and R; later sessions shift progressively towards project development and feedback.

BUSINESS CASE-BASED LEARNING

A Starting Point, Not a Closed Problem

Teams of 3–4 work on a real-world case assigned at the start of the course, developing it progressively as new techniques are introduced. Teams may:

- **Extend** the original business problem
- **Incorporate** additional or publicly available data
- **Generate synthetic data** when necessary
- **Introduce** reasonable assumptions and business requirements
- **Modify** the forecasting horizon or level of aggregation
- **Incorporate** additional variables and evaluate modelling strategies

You are not expected to apply every technique covered in the course. You are expected to determine which techniques are appropriate for your problem — and to justify those decisions.

Every assumption and extension must be documented and justified, methodologically and from a business perspective.

The Final Objective

Not to identify the most sophisticated model — to recommend a forecasting approach that is:

Technically sound

The method suits the data you actually have: its frequency, its length, its patterns, its gaps. You can explain why simpler alternatives were rejected.

Properly validated

Evaluated out of sample, against benchmarks, with residual diagnostics and honest uncertainty — not on the data it was fitted to.

Useful for the decision

It arrives at the right horizon and granularity for the person who must act, and it changes what they would otherwise do.

A well-argued seasonal naïve beats an unexplained neural network. Every time.

TEAM ASSIGNMENTS

Four Cases

CASE 1 · RETAIL & FMCG

Happy Cow Ice Cream

Seven months of point-of-sale data from a single campus store. Which flavours, promotions and locations should a small vegan ice cream maker bet on?

CASE 2 · AIRLINES

Balearic Airlines

One daily Barcelona–Palma flight, 186 seats, two fare classes. How many seats to protect for the higher fare, decided two months out?

CASE 3 · MANUFACTURING

Clean Air Limited

Emission systems for Indian commercial vehicles. Import components with 6–12 month lead times and a full year committed in advance.

CASE 4 · E-COMMERCE AT SCALE

Wayfair

Four million products, monthly demand 1–18 months ahead. Sparse histories, exploding catalogue, mixed seasonality.

CASE 1 · RETAIL & FMCG

Happy Cow Ice Cream

Asia Case Research Centre, The University of Hong Kong · Ref. 18/616C

THE DECISION

The CEO wants to grow sales and is opening more stores. An intern must recommend promotional and assortment actions backed by data, not intuition.

WHAT THE CASE GIVES YOU

A qualitative description of the business, its five customer types, roughly sixteen rotating flavours, differentiated pricing by customer group, and seven months of transaction-level point-of-sale records from the campus store.

WHAT MAKES IT HARD

Seven months is shorter than one seasonal cycle, so annual seasonality cannot be estimated from history alone. Demand is weather- and calendar-driven, and the store manager already intervenes by pushing tastings and swapping flavours.

WHERE YOU WILL HAVE TO EXTEND IT

Define which series you will actually forecast (store revenue, cups per day, flavour group share), reconstruct or simulate the daily series, and bring in Hong Kong weather and the university teaching calendar as external drivers.

CASE 2 · AIRLINES · REVENUE MANAGEMENT

Balearic Airlines

IESE Business School · OIT-34-E

THE DECISION

Set a protection level for priority-class seats — equivalently a booking limit for economy — for each day of the week of 9 October 2023, decided two months before departure.

WHAT THE CASE GIVES YOU

Twelve months of decensorised daily demand for priority-class tickets on the 5 p.m. BCN–PMI flight, a 186-seat aircraft, a two-tier fare structure where priority sells for roughly 50% more, and the National Day holiday falling in the target week.

WHAT MAKES IT HARD

A point forecast is useless here: the decision needs the whole demand distribution, because the trade-off between spoilage and dilution is resolved by a quantile, not a mean. Holidays and day-of-week effects interact.

WHERE YOU WILL HAVE TO EXTEND IT

Decide how to model day-of-week and holiday effects, produce probabilistic forecasts rather than point ones, and connect the forecast quantile to the actual revenue trade-off in euros.

CASE 3 · MANUFACTURING · SUPPLY CHAIN

Clean Air Limited

Ivey Publishing, Ivey Business School · W42478

THE DECISION

Build an inventory and ordering plan for imported components: how much to commit to overseas suppliers a full year ahead, when variations above 5% are penalised and changes need six months notice.

WHAT THE CASE GIVES YOU

Nine imported items across four suppliers in China, Germany and Singapore, with unit values, 6- and 12-month lead times, line-stoppage costs, three years of annual demand and monthly demand history, plus the company profit record.

WHAT MAKES IT HARD

Only three years of monthly history, containing a COVID collapse and a demand surge — trend breaks that mislead most models. Chinese New Year and the Western peak-shipping season distort lead times, not just demand.

WHERE YOU WILL HAVE TO EXTEND IT

Choose the aggregation level (item, supplier, total value), decide how to treat structural breaks, and translate the forecast into an ordering policy where the cost of being wrong is asymmetric and contractual.

CASE 4 · E-COMMERCE · FORECASTING AT SCALE

Wayfair

Foresight: The International Journal of Applied Forecasting · 2025 Q4

THE DECISION

Produce one monthly demand forecast per product, 1 to 18 months ahead, that inventory, labour and transport planning can all treat as a single source of truth.

WHAT THE CASE GIVES YOU

A published account of Demetra, Wayfair's hierarchical engine: a five-level product hierarchy, an ensemble of top-down statistical and bottom-up machine-learning forecasts, differential weighting by horizon and segment, and the reported accuracy gains.

WHAT MAKES IT HARD

Extreme sparsity — three quarters of items have more zero-sales months than non-zero ones and half have under two years of history. Repeated trend breaks, promotions, stockouts masking true demand, and multiple seasonal shapes coexisting.

WHERE YOU WILL HAVE TO EXTEND IT

You will not have Wayfair's data. Build a representative synthetic hierarchy, then test whether hierarchical reconciliation and ensembling actually beat forecasting each series on its own — and at what cost in complexity.

Four Blocks, Four Milestones

Each block ends with something concrete in your project, not just a topic covered.

BLOCK 1 **Sessions 1–4**	**Understanding Time Series** Business problem formulation · Time plots · EDA · Autocorrelation · Trend · Seasonality · Decomposition · Transformations	**CASE MILESTONE** Define the forecasting problem, target, horizon and data requirements. Identify the main temporal characteristics of the problem.
BLOCK 2 **Sessions 5–8**	**Building the Forecasting Baseline** Naive · Seasonal Naive · Simple Exponential Smoothing · Holt · Holt-Winters · ETS · Model evaluation	**CASE MILESTONE** Establish a baseline, develop the first forecasting models and define an appropriate evaluation strategy.
BLOCK 3 **Sessions 10–16**	**Statistical Time Series Modelling** AR · MA · ARMA · Stationarity · Differencing · ARIMA · Seasonal ARIMA · Automatic ARIMA · Complex seasonality	**CASE MILESTONE** Develop and compare more advanced approaches and determine which methodologies are appropriate for your specific business problem.
BLOCK 4 **Sessions 17–18**	**Integration & Business Application** Integrated forecasting workflow · Model comparison · Validation · Business interpretation · Final feedback	**CASE MILESTONE** Complete the technical solution, select the final forecasting approach and translate results into business recommendations.

ASSESSED SESSIONS

Session 9: Intermediate Test (20%) and Checkpoint 1. Session 19: group presentation, part of Group Work (30%). Session 20: individual practical exam in R (40%). Class Participation (10%) is earned across every Case Clinic.

GROUP BUSINESS CASE PROJECT · 30% OF THE FINAL GRADE

How the Project Is Graded

1	**Business problem & data understanding**	Grasp of the business problem, definition of the forecasting target and horizon, identification and preparation of the data, and the reasonableness of assumptions, synthetic data or extensions.	**20%**
2	**Application of forecasting methods**	Correct implementation in R of the techniques seen in class. Using all of them is not what is assessed — choosing and applying the ones that make sense is.	**20%**
3	**Model evaluation & methodological reasoning**	Model comparison, use of baselines and appropriate metrics, validation and diagnostics — and above all the ability to justify why approaches were selected or discarded.	**20%**
4	**Business interpretation & recommendations**	Turning technical results into conclusions that are useful for the original business problem, including limitations and risks.	**20%**
5	**Communication & final presentation**	Clarity, structure, visualisations, ability to synthesise, and defence of the decisions taken.	**20%**

One Rule That Makes the Rest Work

Keep a decision log from day one.

One running document per team. Every week, three lines: what we tried, what we found, what we decided and why. It is the raw material of your final report, and it is what separates a recommendation from a guess.

Bring it to every session. I will read it when I sit down with your team.

A LOG ENTRY, MINIMUM VIABLE

- **Session & date** — and who was there.
- **Assumption or data added** — with its source or the code that generated it.
- **Method tried** — and the accuracy result against the benchmark.
- **Decision** — keep, reject, revisit later, and the reason in one sentence.

TODAY'S SESSION

Getting Started with Forecasting

01 — Why time series analysis matters

02 — What & how to forecast

03 — Data, methods & the forecasting workflow

04 — The statistical perspective & time series patterns

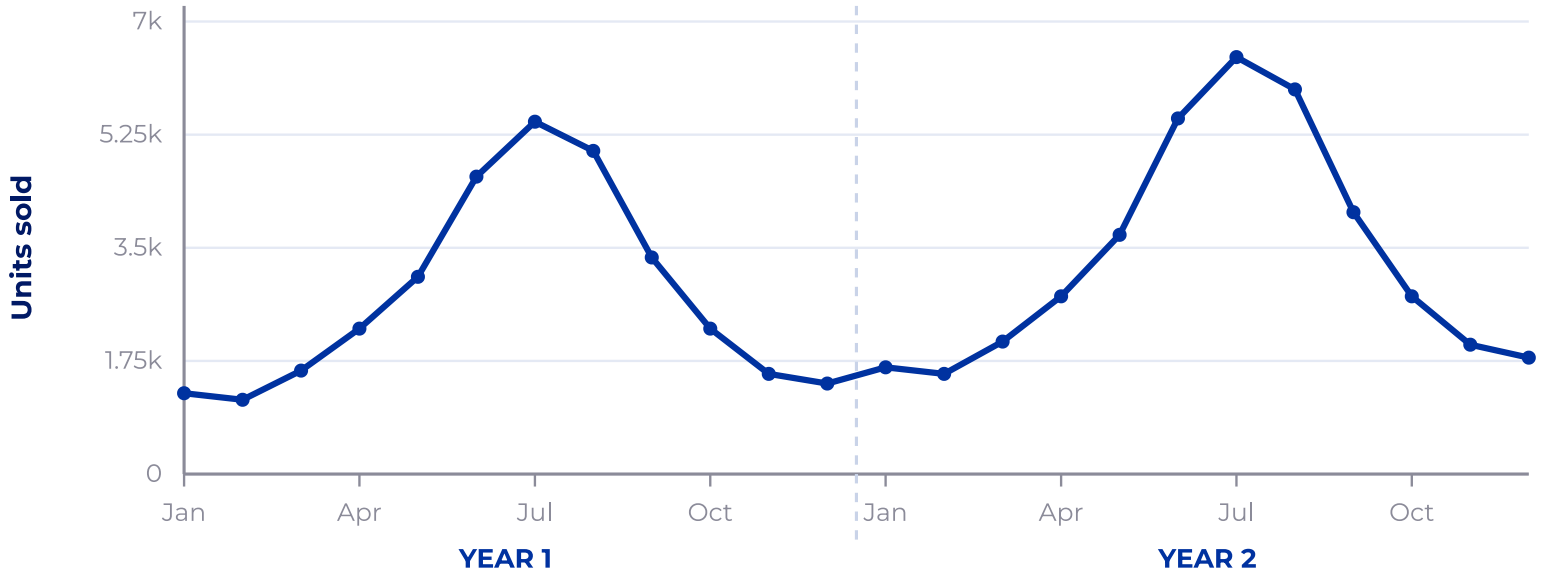
05 — Time series in R: tsibbles & setup

06 — Business case workshop

EXAMPLE THE SERIES WE WILL USE ALL SESSION

Monthly Ice Cream Sales

An illustrative series: 24 monthly observations of units sold by a small ice cream shop. Two things make it a **time series** — the observations are ordered in time, and each one is related to those before it.



Why Time Series Analysis Matters

Any variable observed sequentially over time is a time series — and the **order** of the observations carries information. That single fact is what makes this a separate branch of analytics, with its own theory, its own diagnostics and its own failure modes.

01

The past constrains the future

Successive observations are dependent, not independent draws. Standard regression assumes independence, so ignoring autocorrelation produces intervals that are far too narrow — answers that look confident and are wrong.

02

Business decisions are decisions about the future

Inventory, staffing, capacity, pricing, budgets, procurement. Each has a lead time, and a lead time forces a forecast: the choice is not whether to forecast, but whether to do it explicitly.

03

The uncertainty is the deliverable

These methods quantify how wrong the forecast can be. That is what lets a business set safety stock, protect seats or size a buffer — a single number never could.

THE SAME MATHEMATICS, EVERYWHERE

Healthcare
Heart-rate monitoring, admissions per day

Environment
Hourly temperature, monthly rainfall

Finance
Daily closing prices, volatility

Operations
Call volumes, arrivals, sensors

Business
Sales, demand by product line, bookings

CASE STUDY · REAL-WORLD FORECASTING

Forecasting a National Pharmaceutical Budget

A national government needed to forecast the annual budget for a subsidised pharmaceutical scheme. Total expenditure ran to several billion dollars a year, and had been underestimated by close to a billion dollars in each of the two preceding years.

Producing that budget forecast meant forecasting sales volumes for hundreds of drug groups every month. Almost all of them showed trend and seasonality — and many had sudden jumps caused by subsidy changes or new competitor drugs entering the market.

THE CHALLENGE

One method, applied automatically across hundreds of series — handling trend and seasonality, yet robust to sudden structural change.

Adapted from Hyndman & Athanasopoulos (2021), *Forecasting: Principles and Practice*, §1.5

CASE STUDY · REAL-WORLD FORECASTING

Airline Passenger Demand

A major domestic airline needed weekly passenger forecasts for every route and every fare class, based on six years of history. The data was anything but clean:

- School holidays that didn't align across cities
- A months-long pilots' strike that halted traffic entirely
- Sporting events that moved between host cities
- A low-cost competitor that launched, then folded

Takeaway: real time series rarely look clean. Forecasting methods need to be robust — and their errors need to be evaluated model by model.

Adapted from Hyndman & Athanasopoulos (2021), *Forecasting: Principles and Practice*, §1.5

What Can Be Forecast?

Forecastability is a property of the **problem**, not of the method. Some quantities are close to predictable, others are barely better than a coin toss — and no model recovers information that is not in the data. Judge your case against three conditions before you write a line of code.

01

How well we understand the drivers

If you can name the factors that move the series and say which direction each one pushes, a model can encode them. If demand moves for reasons nobody in the business can articulate, no method will discover them for you.

IN OUR CASES

Happy Cow: temperature, term calendar and tourist flow are nameable. Wayfair: tens of thousands of new products a month, no history and no known driver.

02

How much relevant data exists

Relevant is the operative word. Two years of monthly data is 24 observations and only two winters — too little to estimate seasonality with confidence. History from before a structural break can be worse than no history at all.

IN OUR CASES

Clean Air: three years of monthly demand, one of which is COVID. Happy Cow: seven months, less than a full seasonal cycle.

03

Whether the forecast changes the outcome

If people act on the forecast, the action changes what actually happens, and the model ends up measuring its own effect. Handled deliberately this is fine; unnoticed, it silently corrupts your evaluation.

IN OUR CASES

BalAir: protecting seats caps observed sales, so the recorded data is censored — the analyst had to reconstruct true demand before forecasting it.

Your first task as a team: decide which of the three is your **binding constraint** — and what you will do about it.

Determining What to Forecast

The most expensive mistakes are made before any modelling starts. Step one is to agree, with the people who will use the numbers, exactly what should be predicted — and to accept that most of a forecaster's time goes into locating, cleaning and collating data, not into fitting models.

Granularity — how fine?

Every product, or product families? Every outlet, or regional totals? Finer granularity means more series, sparser data and worse accuracy per series — but some decisions are only actionable at the fine level. Forecast at the level the decision is taken, then reconcile upwards.

Horizon — how far ahead?

Not a preference: it is fixed by the lead time of the decision. BalAir must commit two months before departure, Clean Air a full year, the Happy Cow manager one weekly delivery. The horizon rules out methods before you choose any.

Frequency — how often?

A forecast needed weekly across hundreds of series has to be produced by an automated, reproducible pipeline — which favours automatic model selection over hand-tuning, and makes stability as important as raw accuracy.

Purpose — who acts on it?

Name the role and the decision. It tells you which error is expensive, whether a mean or a high quantile is wanted, and what "good enough" means. A forecast nobody uses is a failed project regardless of its accuracy.

Write one sentence and keep it visible all term: “We forecast [variable] at [granularity], every [frequency], [horizon] ahead, so that [role] can decide [decision].”

Forecasts, Goals and Plans

In business these three words are used interchangeably, and the confusion is a leading cause of bad forecasts. They answer different questions and they must be produced by different processes.

FORECAST

What we think *will* happen

A prediction of the future made as accurately as possible with the information available, including how uncertain it is. It is evidence, so it must be allowed to be unwelcome.

EXAMPLE

Given the last two years and the weather outlook, July sales will be 6,200 units, 80% interval 5,400–7,000.

GOAL

What we *want* to happen

A target. It may be ambitious and it need not be realistic. Nothing is wrong with a goal — the error is letting it overwrite the forecast, or setting it with no forecast to reference.

EXAMPLE

Grow July sales 20% year on year to 7,400 units.

PLAN

What we *do* about the gap

The actions chosen in response to both: what has to change for the goal and the forecast to converge, and what to do if they do not. This is where the forecast earns its keep.

EXAMPLE

Run a campus promotion, add a freezer, and pre-book extra production for weeks 27–30.

The classic failure: a “forecast” that is really the revenue target handed down. Inventory built on it is wrong twice — once in the number, once in the reasoning.

Planning Horizons

The horizon is imposed by the decision, and it dictates what can be modelled: at short horizons the most recent observations dominate; at long ones the level drifts and only structural patterns survive.

HORIZON	TYPICAL DECISIONS	DATA & METHOD	IN OUR CASES
Short days – weeks	Staffing rosters, replenishment orders, production scheduling, daily cash	Daily or weekly data; recent history dominates; simple methods are hard to beat	Happy Cow: the weekly delivery the store manager phones in
Medium weeks – months	Purchasing, budgets, seat and capacity allocation, promotion planning	Monthly or weekly data; seasonality matters most; ETS and ARIMA territory	BalAir: protection levels set two months before the flight
Long quarters – years	Strategy, plant and contract commitments, technology investment	Aggregated data; scenarios and judgment alongside models; wide intervals are honest	Clean Air: the full-year supplier schedule confirmed in advance

Same series, three different problems. Before choosing a model, be explicit about which horizon your recommendation serves.

Qualitative and Quantitative Forecasting

Which family applies is decided by the data, not by taste. Most real engagements use both — and the difference between a consultancy and a guess is knowing which parts of the answer came from which.

Qualitative

Used when there is no relevant history: a new product, a new market, a regulatory change, a structural break that makes the past uninformative. It is not guesswork — it is structured judgment: Delphi rounds, expert panels, analogous-product analysis, scenario building, market research.

Watch out for anchoring on the last number seen, optimism bias, and the loudest voice in the room. Structure the process and record the reasoning.

Quantitative

Applies when two conditions hold: numerical information about the past exists, and it is reasonable to assume that some of those patterns will persist. Everything in this course — benchmarks, exponential smoothing, ARIMA — lives here.

Watch out for the second condition. When the mechanism changes, a well-fitted model keeps extrapolating a world that no longer exists.

In your case both will appear: judgment sets the assumptions and the synthetic data, models produce the forecast, judgment interprets it. Label each one in your log.

Three Types of Forecasting Models

Take hourly electricity demand (ED) in a hot region in summer. The same quantity can be modelled three ways, and the choice is not about sophistication — it is about what you will know at the moment you have to forecast.

Explanatory

`ED = f(temperature, economy, population, time of day, day of week, error)`

Explains **why** demand varies, and supports what-if analysis: what happens if next summer is 2°C hotter?

Cost: to forecast, you must first forecast every predictor. Temperature six months out is harder than demand.

Time series

`ED[t+1] = f(ED[t], ED[t−1], ED[t−2], …, error)`

Uses only the variable's own past. It does not explain the mechanism, but it needs nothing you do not already have.

The focus of this course — and usually the most accurate option at short and medium horizons.

Mixed (dynamic regression)

`ED[t+1] = f(ED[t], temperature, time of day, day of week, error)`

Past values plus external predictors: the pragmatic choice when a driver is known **and** knowable in advance, like a holiday calendar or a planned promotion.

Cost: more parameters, more ways to be wrong.

Ask of every candidate driver: will I actually know its value at forecast time? If not, it belongs in the explanation, not in the model.

The Forecasting Workflow: Five Steps

Every forecasting project follows the same five steps, and each one produces an artefact you can show. Your team's work this term is exactly this loop, run once for your case.

1	**Problem definition**	Who needs what, at which granularity and horizon, and what a wrong answer costs	→ the one-sentence brief
2	**Gathering information**	Statistical data plus the accumulated expertise of the people who produce and use it	→ a clean tsibble + assumptions log
3	**Exploratory analysis**	Graph everything: trend, seasonality, cycles, outliers, changing variance	→ plots + a list of features to model
4	**Choosing & fitting models**	Two or three candidates against a benchmark, estimated on training data only	→ a comparison table
5	**Using & evaluating**	Accuracy out of sample, residual diagnostics, and the organisational work of acting on it	→ the recommendation

STEPS 1 & 2

Defining the Problem, Gathering the Information

These two steps decide whether the project can succeed at all, and they are where consultants earn their fee. Neither involves a model.

1 · Problem definition

Usually the hardest part: it requires understanding how the forecast will be used, by whom, and how the forecasting function sits inside the organisation. Talk to everyone involved before building anything.

- Which decision does this forecast feed, and when is it taken?
- Which error costs more — too high or too low, and by how much?
- What happens today without a forecast? That is your true benchmark.
- What would make the client stop using it? Late, too granular, unexplainable.

2 · Gathering information

Two kinds are always needed: the statistical data, and the accumulated expertise of the people who collect it and will act on it. The second kind is the one nobody writes down.

- Do not discard old history without reason — a good model absorbs gradual change.
- Check for definition changes: a series that switched units or scope mid-way is two series.
- Distinguish **zero sales** from **no data**, and demand from sales when stockouts exist.
- Collect the calendars: holidays, promotions, launches, plant shutdowns.

Rule of thumb from practice: expect to spend more time on steps 1–2 than on steps 4–5. Budget for it in your project.

STEP 3

Exploratory Analysis: Always Graph the Data First

A plot answers in ten seconds what a table hides for hours. Each question below has a direct consequence for the model you will be allowed to use — this is not decoration, it is model selection starting early.

Is there a trend?

A sustained long-term movement, not necessarily linear or monotonic.

Consequence: a flat-mean method is disqualified; you need a trended model or differencing.

Is seasonality important?

A pattern of known, fixed period — week, month, quarter, year, or several at once.

Consequence: the benchmark becomes seasonal naive, and seasonal terms enter the model.

Are there business cycles?

Rises and falls with no fixed period, usually longer than two years.

Consequence: never model a cycle as seasonality — and be humble about long horizons.

Outliers, breaks, changing variance?

A strike, a promotion, a pandemic, a series whose swings grow with its level.

Consequence: explain each one, then decide: keep, adjust, transform, or split the sample.

The graphical toolkit for all four questions — time plots, seasonal and subseries plots, ACF — arrives in Sessions 2 and 3.

STEPS 4 & 5

Fitting Models, and Finding Out Whether They Work

A model that fits the past well is not a good model. The only evidence that counts is performance on data the model has never seen.

4 · Choosing & fitting

Which model is appropriate depends on the data available, the patterns you found in step 3, and how the forecast will be used. Always fit a set of candidates, never a single favourite.

- Start with a benchmark: naive, seasonal naive, or the client's current method.
- Add two or three real candidates — ETS, ARIMA, dynamic regression.
- Estimate parameters on a training set only; keep the last stretch untouched.
- More parameters is not more skill: complexity has to buy accuracy.

5 · Using & evaluating

True performance is only known once the future arrives, so we simulate that: hold data back, forecast it, measure the error, and inspect what the model failed to capture.

- Accuracy measured on the test set, at the horizon the business actually uses.
- Residual diagnostics: what is left over should look like noise.
- Report intervals, not just a number — and check they are calibrated.
- Then the organisational half: who reads it, how often it is refreshed, when to re-fit.

Sessions 5 and 6 give you the benchmarks, the validation scheme and the accuracy measures. Your Checkpoint 1 depends on them.

The Statistical Forecasting Perspective

A future value is not a number we do not know yet — it is a **random variable**. Everything we produce is a statement about its distribution, conditional on everything observed so far. Getting this right is what separates a forecast from a number.

Forecast distribution

`y[t+h] | I[t]`

The set of values the future observation could take, each with a probability, given the information set available today. This is the complete forecast; everything else is a summary of it.

In our cases: BalAir needs the whole distribution — a mean tells Carmen nothing about spoilage versus dilution.

Point forecast

`ŷ[t+h|t] = E( y[t+h] | I[t] )`

The mean of that distribution — the value that minimises squared error. By construction, actual demand lands above it roughly half the time. Plan to it only when over and under cost the same.

In our cases: Clean Air ordering to the mean means running out in half the months, at ₹75,000 per line stoppage.

Prediction interval

`80% or 95% coverage`

A range expected to contain the true value with a stated probability. It widens with the horizon, because uncertainty compounds — a forecast that does not widen is hiding something.

In our cases: Wayfair converts intervals into safety-stock levels per item and region.

Practical consequence: the business question is rarely “what is the mean?” It is “which quantile do I plan to?” — and that depends on the cost of each kind of error.

EXAMPLE THE SAME ICE CREAM SERIES, SIX MONTHS FORWARD

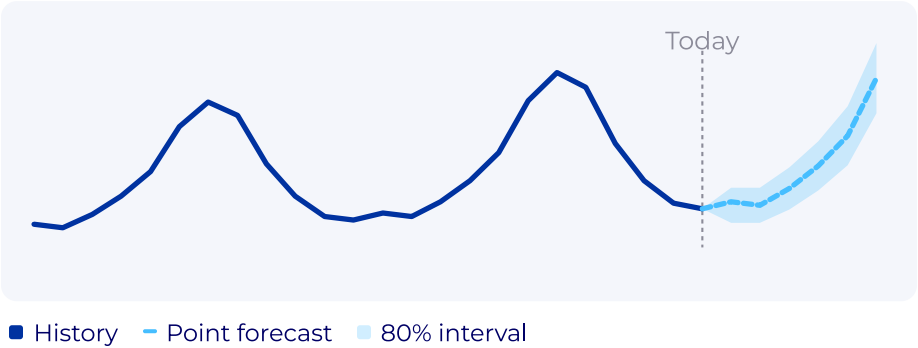
A Forecast Is a Fan, Not a Line

The dashed line continues the trend and the seasonal shape the model found in the history. The shaded fan is where the truth is expected to fall.

An **80% interval** is narrow and will be wrong one month in five. A **95% interval** is wide and almost always right. Neither is “better”: you pick the coverage that matches what an error costs.

Notice the fan opening up as the horizon grows. That widening is information: it tells the client how far ahead the forecast is worth acting on.

In your case: a stockout costs more than surplus → plan to the upper bound. Idle capacity costs more than a queue → plan to the lower. The interval is what makes that choice possible.



The Three Patterns You Are Looking For

Most series are a combination of these three plus noise. Naming them correctly is the whole point of exploratory analysis — confusing a cycle with seasonality is one of the most common and most expensive mistakes in forecasting.



Trend

A long-term increase or decrease in the level. It does not have to be linear, and it can change direction: our shop sells more in year 2 than in year 1 across every month.

Why it matters: a method that forecasts a flat mean will drift below the truth for ever.



Seasonality

A pattern with a **known, fixed period** tied to the calendar or the clock: July peaks and January troughs, Monday spikes, evening rush hours. A series can carry several at once.

Why it matters: it is predictable, so the bar rises: your benchmark becomes seasonal naive, not naive.



Cyclic

Rises and falls with **no fixed period**, usually lasting more than two years and larger in amplitude than seasonality: investment cycles, the Indian commercial-vehicle cycle in the Clean Air case.

Why it matters: its length is unknown, so it cannot be extrapolated safely — say so instead of pretending.

Session 4 gives you decomposition: the formal tool that separates a series into trend, seasonal and remainder so each can be judged on its own.

Time Series Data in R: the tsibble

A **tsibble** is a "time series tibble": a data frame that knows which column is time and which columns identify separate series. Everything in this course starts by getting your data into one — including your case.

- **Index** — the time column that orders the observations. Its type sets the frequency: `yearmonth()`, `yearquarter()`, `yearweek()`, a `Date` or a timestamp.
- **Key** — the column(s) that distinguish multiple series in the same table: store, route, SKU, queue. One row per key per time point, and no duplicates allowed.
- **Measured variables** — everything else: sales, price, temperature, promotion flag.
- Built on the **tidyverse**, so `filter()`, `mutate()`, `group_by()` and `summarise()` all work — but respect time.
- Gaps are explicit: `has_gaps()` and `fill_gaps()` reveal missing periods that a plain data frame silently hides.

```
library(fpp3)

sales <- read_csv("sales.csv") |>
  mutate(month = yearmonth(month)) |>
  as_tsibble(index = month, key = store)

sales
#> A tsibble: 48 x 3 [1M]
#> Key:  store [2]
#>    month  store  sales
#> 2024 Jan  A      1150
#> 2024 Feb  A      1100
```

The `[1M]` is the detected interval — monthly. If it prints as something you did not expect, your index type is wrong.

First R task for your team: get your case data into a tsibble and print it. If it will not build, the problem is in the data — and finding that out now is the point.

Setting Up R

Either editor works for this course. What matters is that `library(fpp3)` loads cleanly before Session 2 — the Business Case Lab time is for your case, not for installations.

Option A · RStudio

The path of least resistance, and what most of the class will use.

- Install **R** from CRAN, then **RStudio Desktop** (free).
- Install the **tidyverse** and **fpp3** packages — the second pulls in tsibble, fable and feasts.
- Create a project folder per team so paths and data travel with the work.

```
install.packages("tidyverse")
install.packages("fpp3")
```

Option B · VS Code + R extension

More setup, but familiar if you already live in VS Code.

- Install **R** (≥ 3.4.0), then the **R extension** (REditorSupport.r).
- In R, install `languageserver` for completion, linting and debugging.
- Optional but pleasant: **radian** as console and **httpgd** as plot viewer.

```
install.packages("languageserver")
## Command Palette →
## "R: Create R Terminal"
```

CHECK IT WORKS

```
library(fpp3); aus_production |>
autoplot(Beer)
```

A plot appears → you are ready. An error → bring the message to Session 2.

HOMEWORK

Before Session 2

Your case

- Read your business case in full, individually, before you meet as a team.
- Open the team **decision log** and write the one-sentence brief: variable, granularity, frequency, horizon, decision-maker, decision.
- List the data you have, the data you could find in public, and the data you will have to generate.
- Name one thing you would ask the client if you could send a single email.

Method

- Read **FPP3 chapters 1 and 2.1** — otexts.com/fpp3, free online.
- Complete the “Time series in R: tibbles and tsibbles” exercise set.
- Review the solved exercises on the course platform before class, not after.
- Bring one plot of any series that interests you — produced by you, in R.

Toolchain

Arrive with R, your editor and `fpp3` installed and verified. Session 2 starts with your own data on screen.

If the install fails, post the error on the forum before class so we can fix it in minutes rather than in the lab.

Session 2 is not a lecture you attend — it is the second working session on your case. Come with the brief written.

TEAM WORKSHOP · 30 MIN

Scope Your Case

- Form your team of 3–4 and collect your assigned case.
- Read it fast — you are looking for the decision, not the details.
- Answer the five questions on the next slide. One slide, bullet points, no prose.
- Three minutes per team to present. Disagreement between teams is the point.

DELIVERABLE · SESSION 1

Five Questions

1. **What exactly will you forecast?** Name the variable, the frequency, the granularity and the horizon. "Demand" is not an answer; "weekly cups sold per flavour group, 12 weeks ahead" is.

2. **Whose decision does it serve, and when do they need it?** Name the role, the decision, and the lead time that fixes your horizon.

3. **How forecastable is it?** Test the three conditions: do we understand the drivers, is there enough data, and does the forecast itself change the outcome?

4. **What data will you have?** Split it three ways: given by the case, obtainable in public, and to be synthesised. For the third, state the assumptions you will simulate from.

5. **What would make the forecast useful?** Which error is more expensive, over or under? What accuracy would beat what the business does today?

WHAT A STRONG ANSWER LOOKS LIKE · CASE 1, HAPPY COW

Specific, Bounded, Justified

1 · What. Daily cups sold at the campus store, split into three flavour groups (core, fruity, warming), four weeks ahead.

2 · For whom. The store manager, who phones in one weekly delivery and holds about one week of freezer stock. The lead time is the horizon.

3 · Forecastability. Drivers well understood (weather, term calendar, tourists); data thin — seven months, under one annual cycle; and the manager's tastings mean the forecast partly influences its own outcome.

4 · Data. *Given:* POS transactions with price, product and timestamp. *Public:* Hong Kong daily temperature, HKU teaching calendar, public holidays. *Synthetic:* extend to 24 months using the observed weekly shape plus a temperature-driven annual cycle — stated as an assumption, not as fact.

5 · Usefulness. A stockout of a core flavour loses a sale and a regular customer; surplus only occupies freezer space, the true bottleneck. So plan core flavours to a high quantile, rotating flavours to the mean. Beat the manager's judgement, which is the current method.

Note what this answer does **not** do: name a model. That comes after the data tells you what patterns exist.

SESSION 1 RECAP

Key Takeaways

Forecastability depends on understanding, data, and whether the forecast changes the outcome

A forecasting task follows five steps, from problem definition to evaluation

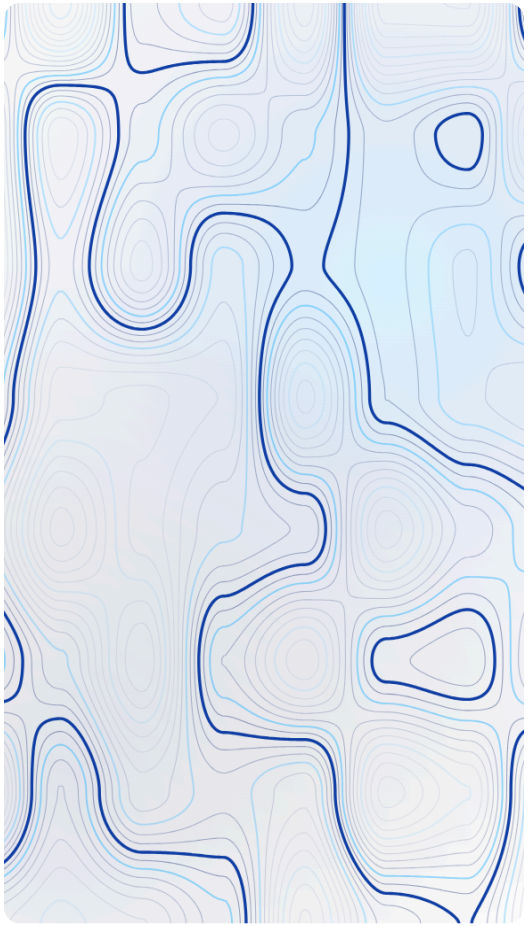
tsibbles are the data structure we'll use in R throughout the course

Next session: time series graphics — time plots, seasonal plots and subseries plots

Forecasts, goals and plans are distinct — don't confuse them

A forecast is a distribution — point forecasts need prediction intervals

Your team owns one real case for the whole course — you choose which techniques apply, and justify it



END OF SESSION 01

TIME SERIES ANALYSIS

Introduction & Course Setup

NEXT · SESSION 2 · 03 SEPTEMBER

Time series graphics

Time plots, seasonal plots, subseries plots and scatterplots — the four graphics that decide which models you are allowed to use.

What follows is annex material: the full 20-session plan, the grading breakdown and the project rubric. Reference only — not projected in class.

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